CATHOLIC JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE NAME			
CLASS	2T	**INDEX NUMBER**	

BIOLOGY

9648/02
23^{rd} August 2016
2 hours

Additional Materials: Writing Paper

READ THESE INSTRUCTIONS FIRST

Write your index number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper. **[PILOT FRIXION ERASABLE PENS ARE NOT ALLOWED]**
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

There are two sections in this paper.

Section A]
Answer all questions

Section B]
Answer all questions. Answer each part on a **separate** piece of paper.

At the end of the examination, fasten all work securely together.
The number of marks is given in brackets [ ] at the end of each question or part of the question.

For Examiner's Use	
Section A	**80**
1 [10]	
2 [10]	
3 [10]	
4 [10]	
5 [10]	
6 [10]	
7 [10]	
8 [10]	
Section B	**20**
9a/10a	
9b/10b	
9c/10c	
TOTAL	**100**

This document consists of 21 printed pages and 1 blank page.

[Turn over

Section A

Answer **all** questions in this section.

1 The figure below shows three structural components of the membrane system in a pancreatic cell.

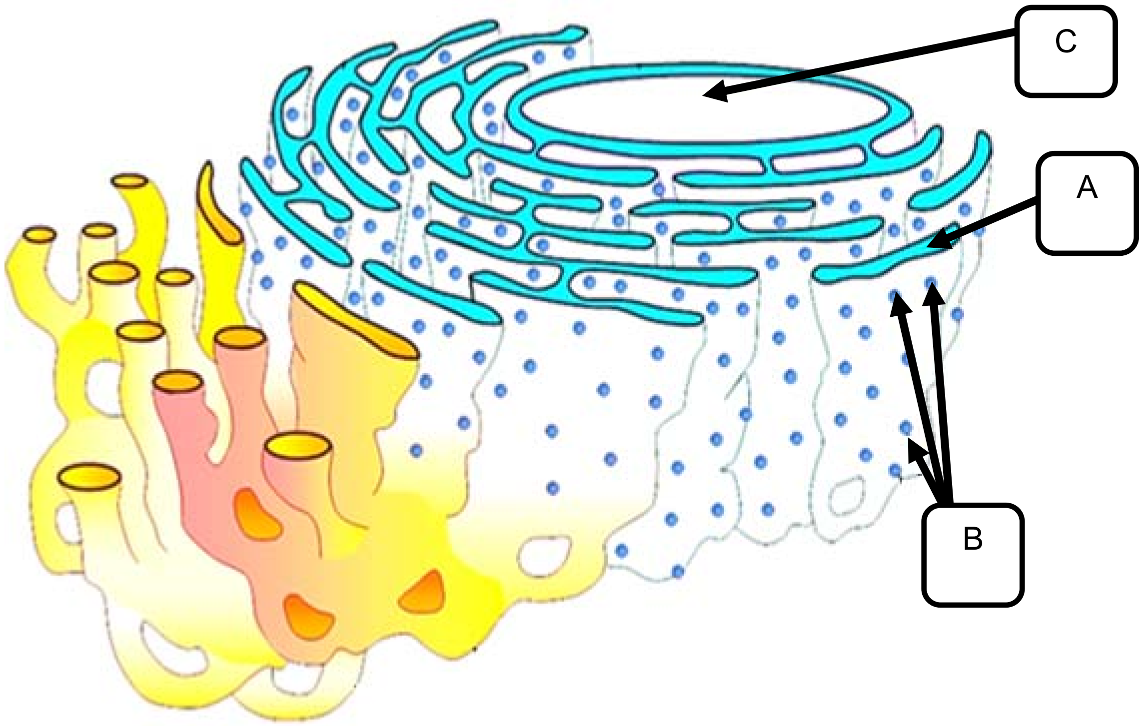

Fig. 1.1

(a) State the name and function of the structures in Fig. 1.1.

Structure A

Name: ………………………………………………………

Function: ……………………………………………………… [1]

Structure B

Name: ………………………………………………………

Function: ……………………………………………………… [1]

(b) State what is **structure C** and explain how **structure A** remains in close proximity to **C.**

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………[2]

Fig. 1.2 shows a hormone essential for regulation of high blood glucose levels.

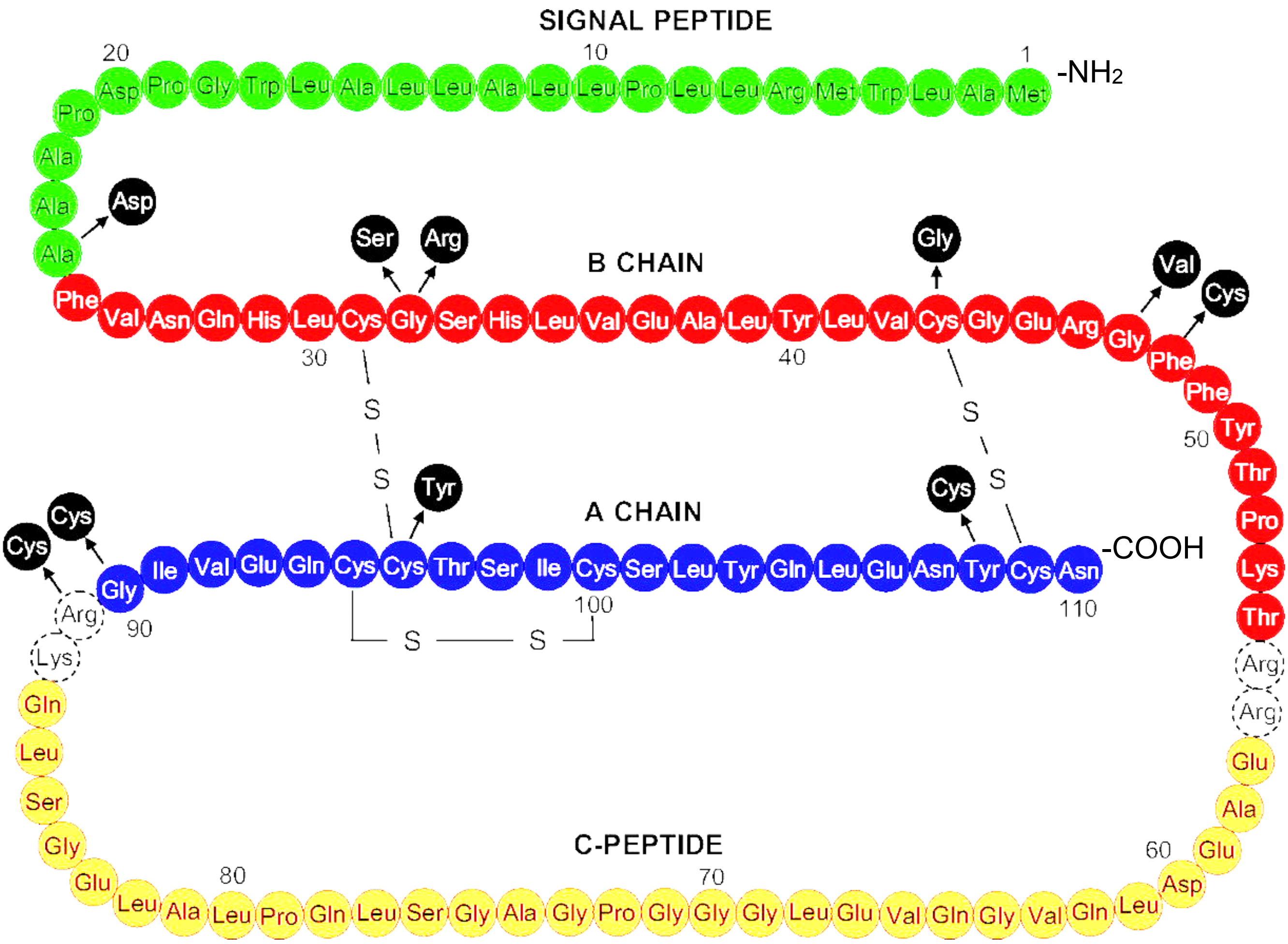

Fig. 1.2 Unit sequences of wild type (WT) and the mutant hormone. The mutations are noted in black circles together with location in the B (25 to 5) or A chain (90-110). The dashed circles indicate the basic residues that are activation cleavage sites.

(c) With reference to Fig. 1.2, draw out the bond (ensure correct orientation) formed between units 109 and 110 of the A-chain.

[2]

(d) With reference to Fig 1.2, suggest how amino acid substitutions between 90 and 109 could occur and what would be the corresponding effect on the resultant mutant hormone.

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………[4]

[Total: 10]

2 The figure shows the molecular configuration of chymotrypsin. Chymotrypsin is one of the major proteases in the human digestive tract, in which its role is to hydrolyse large protein molecules into smaller peptides that are then further processed by peptidases. Fig. 2.1a shows a blown up representation of a portion of chymotrypsin shown in Fig. 2.1b.

Fig. 2.1a

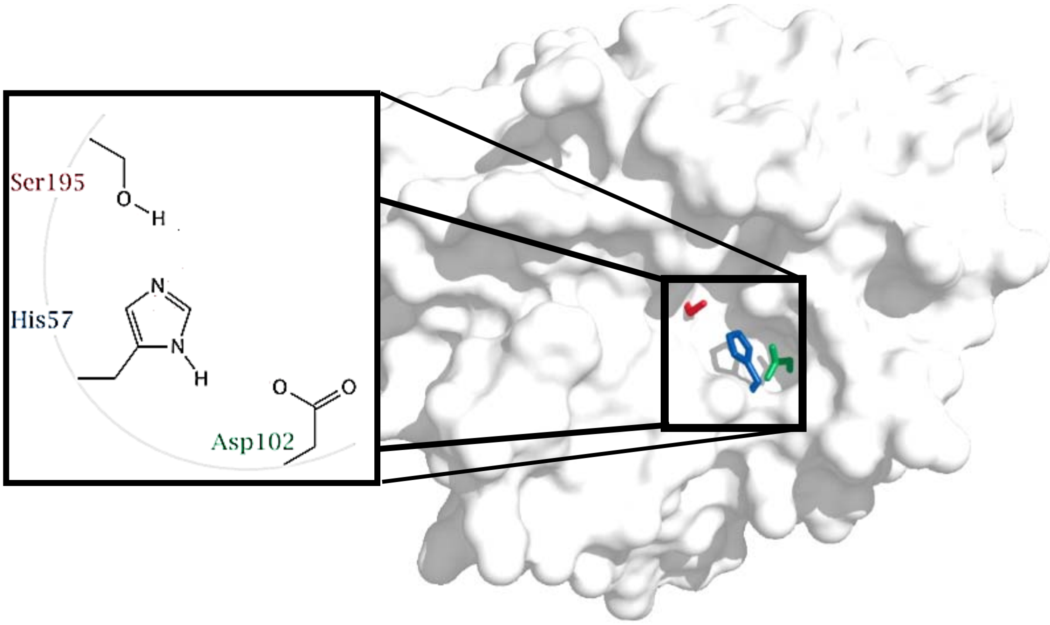

Fig. 2.1b

(a) Using the ‘induced-fit hypothesis’, explain the mechanism in which chymotrypsin carries out its function.

....

....[2]

Fig. 2.2 shows the energy exchange in a chemical reaction without the enzyme present.

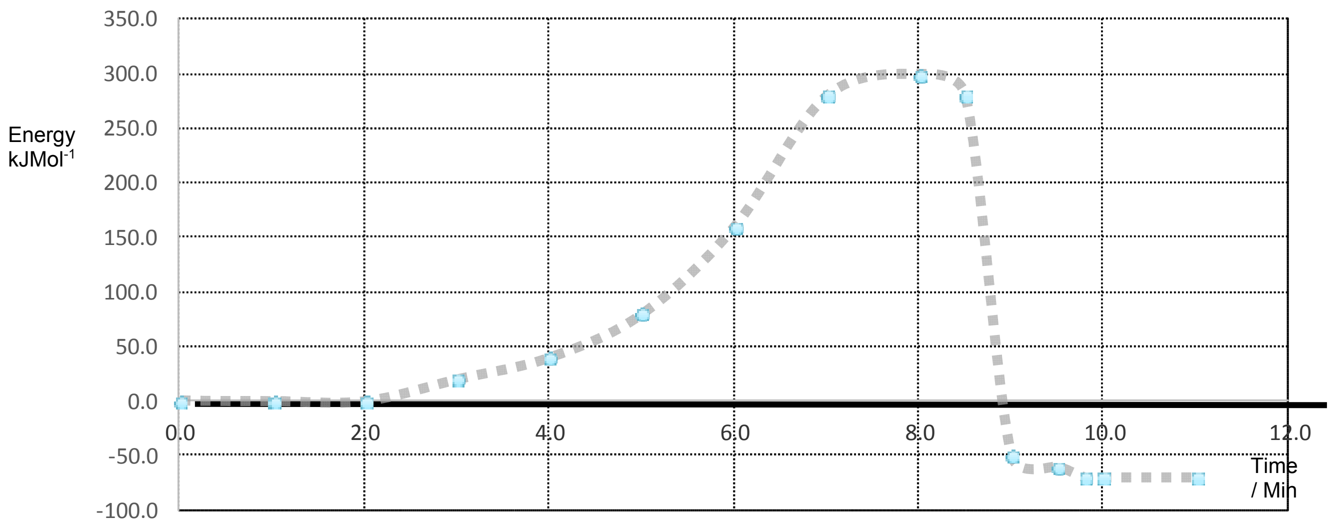

Fig. 2.2

(b) On the graph above draw out the plot tracing the effect of enzyme action and label the effect of enzyme on activation energy. [2]

Fig. 2.3 shows the effects of substrate on enzyme reaction.

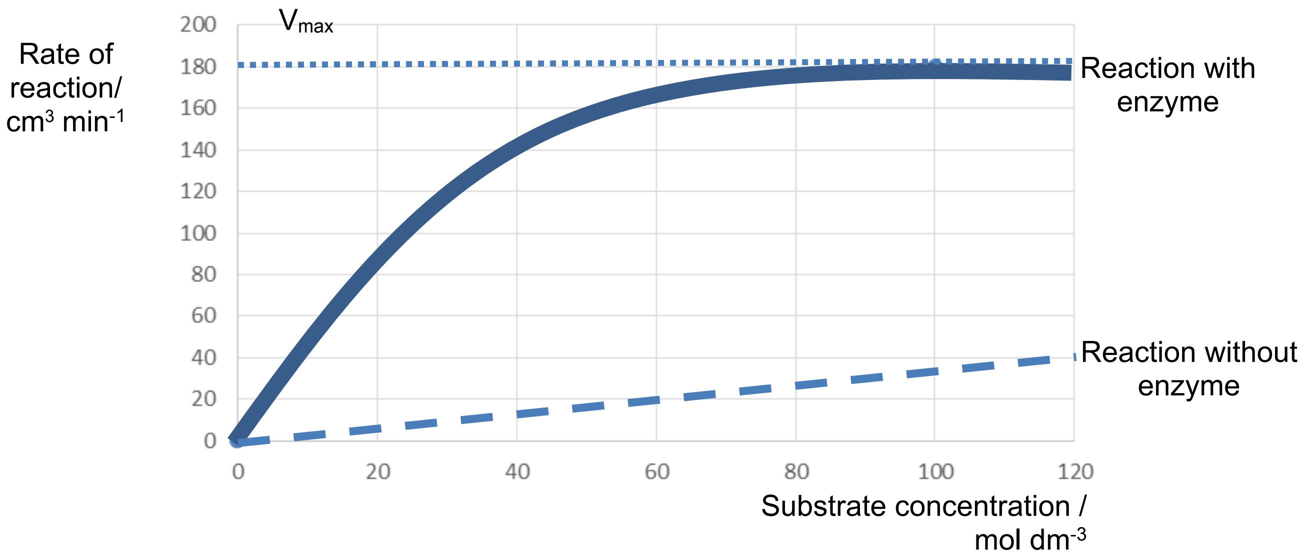

Fig. 2.3

(c) Explain what can be inferred from the graph in Fig. 2.3, with reference to substrate concentration and limiting factors.

....

....

....[2]

(d) On Fig. 2.3 draw a curve representing the effect of a non-competitive inhibitor on rate of reaction with no change in the affinity of the enzyme.

[2]

(e) Account for why the Michaelis constant for both the non-competitive inhibitor and the reaction without inhibitor is the same.

....

....

....[2]

[Total: 10]

3 Cells were transferred and grown in ^{15}N medium for many generations before they were transferred to ^{14}N medium again and allowed to divide.

DNA was extracted periodically from the culture and subjected to density gradient centrifugation using caesium chloride.

Fig. 3.1 shows the density gradient results across three generations.

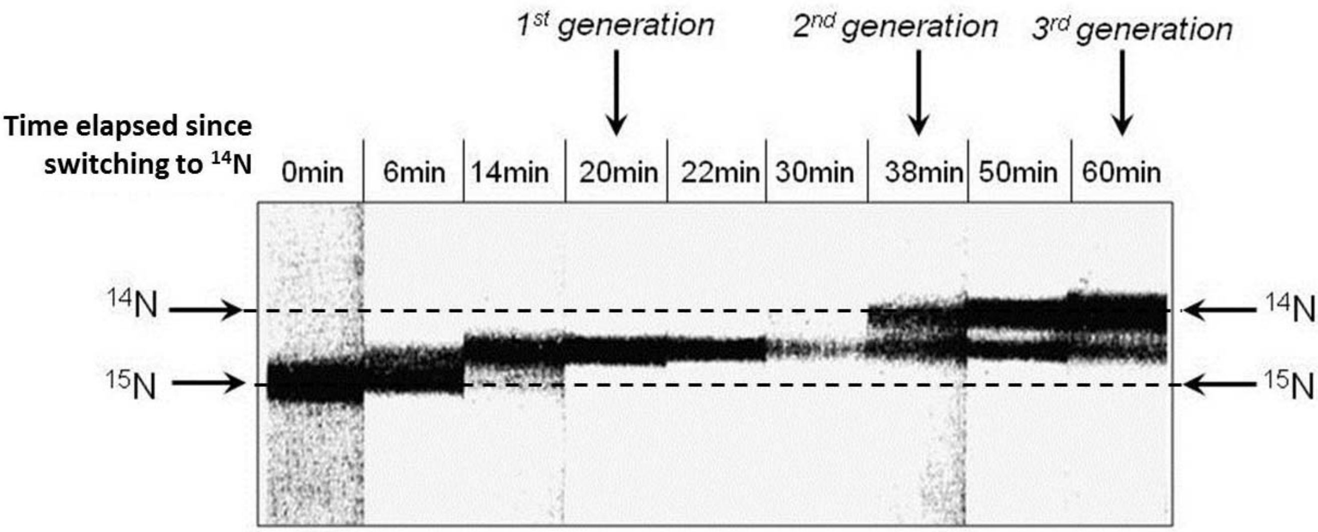

Fig. 3.1

(a) With reference to Fig. 3.1, account for the model of DNA replication which these cells undergo.

………

………

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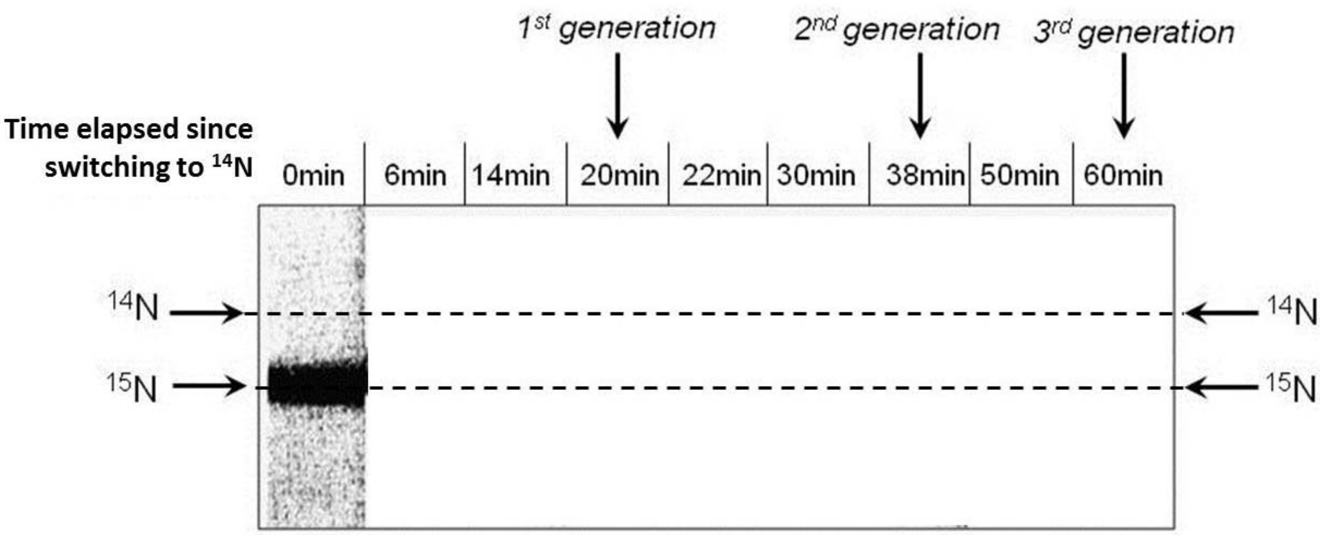
……....[3]

(b) State another model of DNA replication not shown in Fig. 3.1 draw only its band patterns for the 1st, 2nd and 3rd generations in the Figure below.

Model of DNA replication: …………………………………………….



[2]

Fig. 3.2 shows a simplified representation of DNA replication occurring on a linear chromosome.

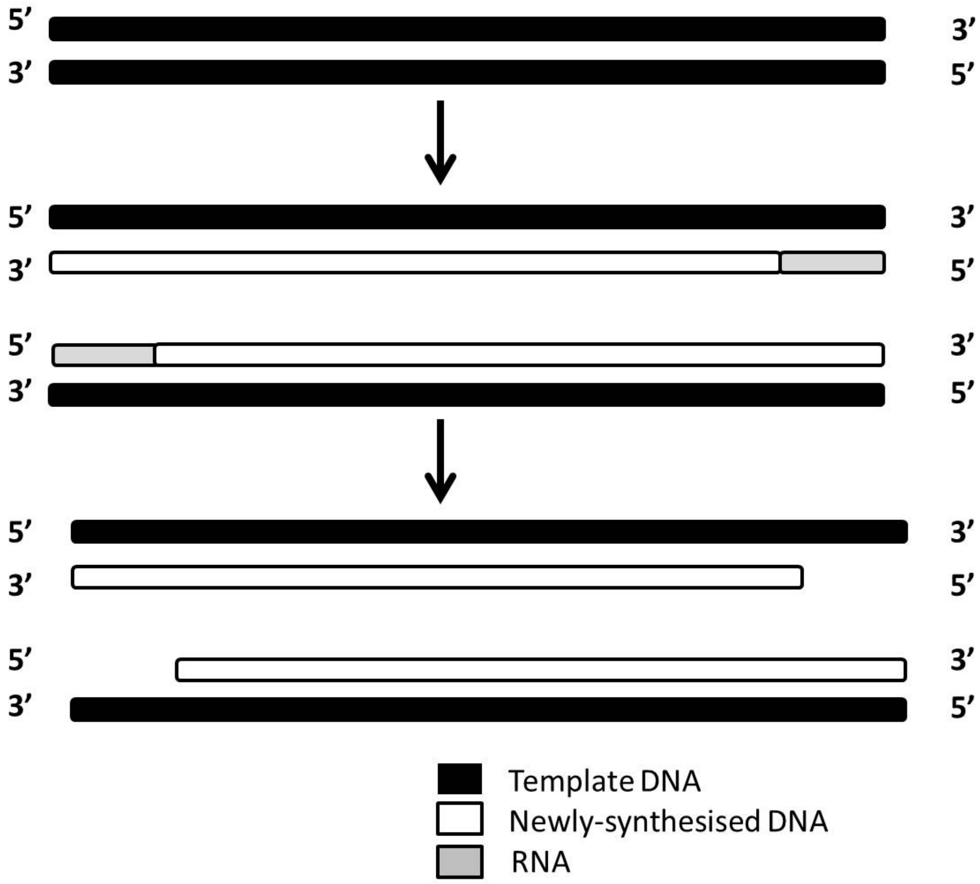


Fig. 3.2

(c) Explain how certain cells address the molecular issue reflected in Fig. 3.2.

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……

………………………………………………………………………………………………………...[2]

The cellular process shown in Fig. 3.2 has many similarities with translation even though the products formed are different. Some of the similarities are that they both take place in 3 different stages (initiation, elongation, termination), both require energy, monomers for extension, a template for product synthesis as well as bond formation involving the removal of a water molecule.

(d) State three other similarities between the cellular process shown in Fig. 3.2 and translation.

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…… [3]

[Total: 10]

4 Fig. 4.1 shows two different stages, A and B (as shown by arrow) of the HIV reproductive cycle.

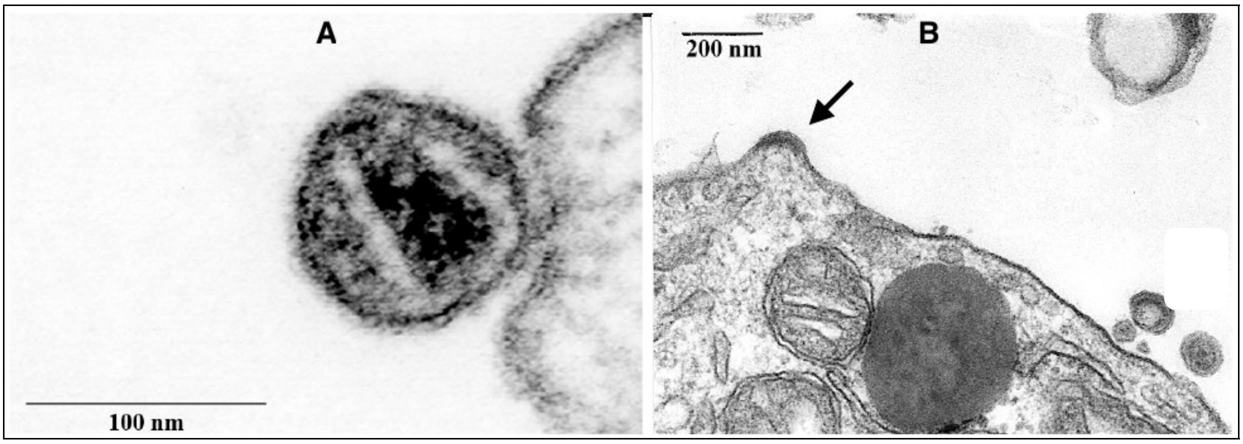

Fig. 4.1

(a) Describe the events occurring in stage A of the HIV reproductive cycle.

....

....

....[2]

(b) Compare the stage immediately following stage A with stage B.

....

....

....[2]

(c) Contrast the final stage of the reproductive cycle of HIV and T4 phage.

....

....[1]

For treatment, HIV-infected patients can receive HIV antiviral drugs such as Tenofovir (Fig. 4.2A).

Fig 4.1B shows an adenosine triphosphate (dATP) molecule, which shares similar chemical groups to Tenofovir.

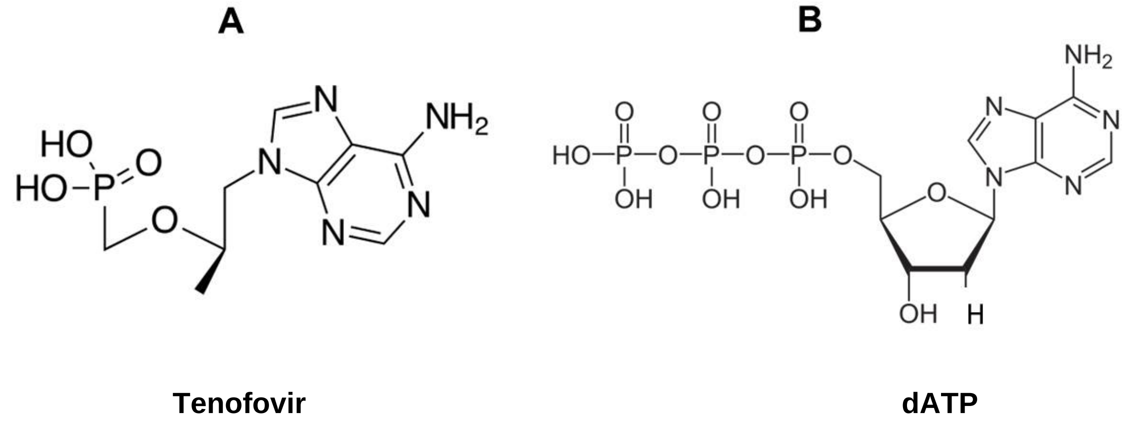

Fig. 4.2

(d) With reference to Fig. 4.2, suggest how Tenofovir acts as a drug that interferes with the HIV reproductive cycle.

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……...[2]

HIV entry into cells requires involvement of at least one type of co-receptor. CCR5 is required for HIV virus entry. CCR5 Δ32 is a 32-base-pair deletion that introduces a premature stop codon into the CCR5 receptor locus, resulting in a non-functional receptor.

Timothy Ray Brown was an AIDS patient who received a hematopoietic stem cell transplant from a donor with homozygous CCR5 Δ32 on the CCR5 gene. After the transplant, he stopped his antiretroviral treatment. Following that, it was found that Timothy's HIV viral levels steadily decreased and his CD4 T- cell count increased. Eventually, he was found to be cured from HIV.

(e) Suggest why not all HIV-infected patients can be cured with this therapeutic method.

………

……...[1]

(f) Explain how HIV infection may result in the death of infected CD4 cells.

………

………

……...[2]

[Total: 10]

5 Fig. 5.1 shows the formation of a eukaryotic transcription initiation complex.

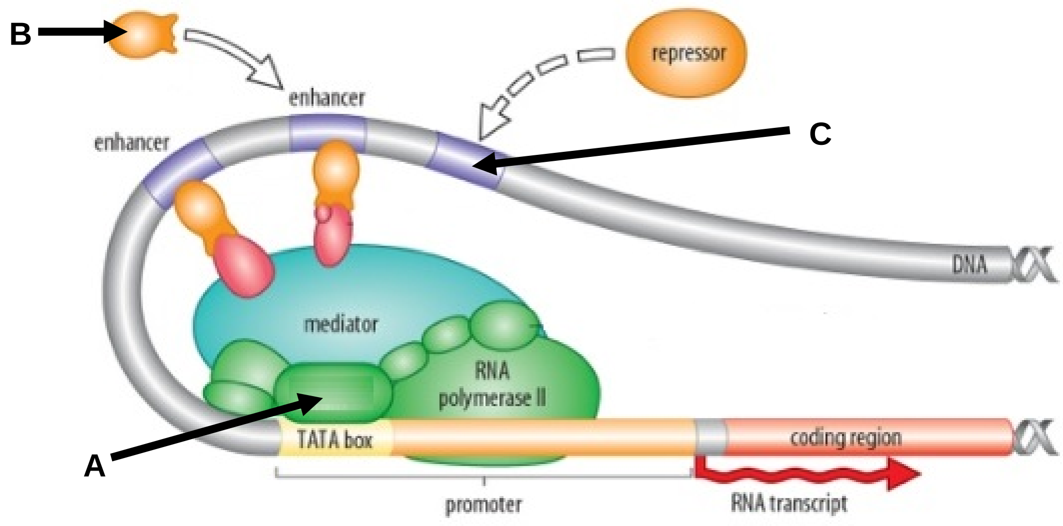

Fig. 5.1

(a) Identify the following proteins.

A: ______________________________________

B: ______________________________________

C: ______________________________________

[3]

(b) Explain how the eukaryotic transcription initiation complex for high rate of transcription can be formed.

....

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....

....[4]

High rate of transcription caused by mutations in cancer critical genes may result in dysregulation of cell cycle control and subsequently lead to uncontrolled cell division.

c-myc is a regulator gene that codes for a transcription factor. A mutated version of c-myc that is highly expressed is found in many cancers. Fig 5.2 shows the relative level of expression of c-myc in normal and cancer prostate tissue.

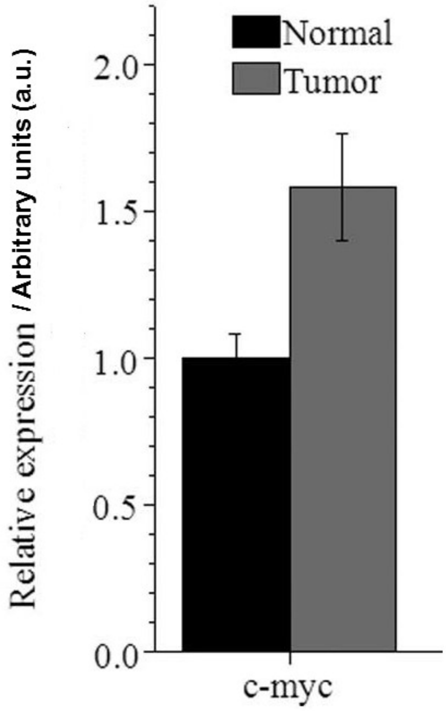

Fig 5.2

(c) With reference to Fig.5.2, explain how mutation in the c-myc gene led to increased expression in cancer prostate tissue.

………

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………[3]

[Total: 10]

6 Fig. 6.1 shows a Calico cat with a mosaic coat with patches of orange and black. It is known that fur coat colour in cats is determined by a single gene. Only female cats can develop calico fur coat. Male cats usually have only orange or black fur coat.

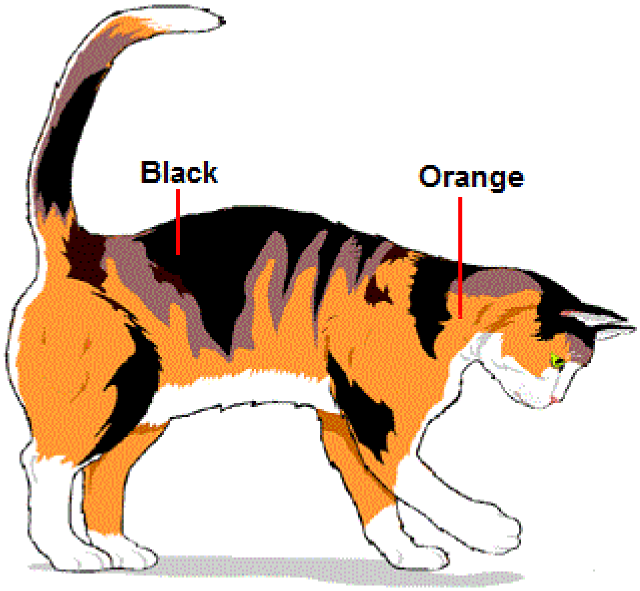

Fig. 6.1

(a) Identify the type(s) of inheritance determining Calico fur coat colour in cats.

....

....[1]

(b) Using B to represent allele for black coat and R to represent allele for orange coat, draw a genetic diagram to show how a cat-breeder can obtain Calico cat from a cross between a pure-breeding black male and an orange female cat.

[4]

Coat colour inheritance in horses is different from cats. Two unlinked genes *E* and *G* control coloured coat in horses. The two genes are thought to be involved in the same metabolic pathway for pigment formation.

- Horses may be bay, black or chestnut in colour.
- Horses may be bay / black when at least one dominant allele *E* is present.
- Chestnut coat colour is always produced in the presence of two copies of the e allele.

Horses coat colour goes through a natural graying process. Horses born with bay, black or chestnut coat colour will steadily turn gray. This process is mediated by a single copy of the dominant allele G regardless of the genotype of the gene E controlling coat colour.

(c) Draw a genetic diagram in the space below to show the result of the cross between two gray horses that were heterozygous at both gene loci G/g and E/e and the resultant phenotypic ratio of the offspring.

[5]

[Total: 10]

7 Fig. 7.1 shows the different modes of signal conduction in a non-myelinated neuron.

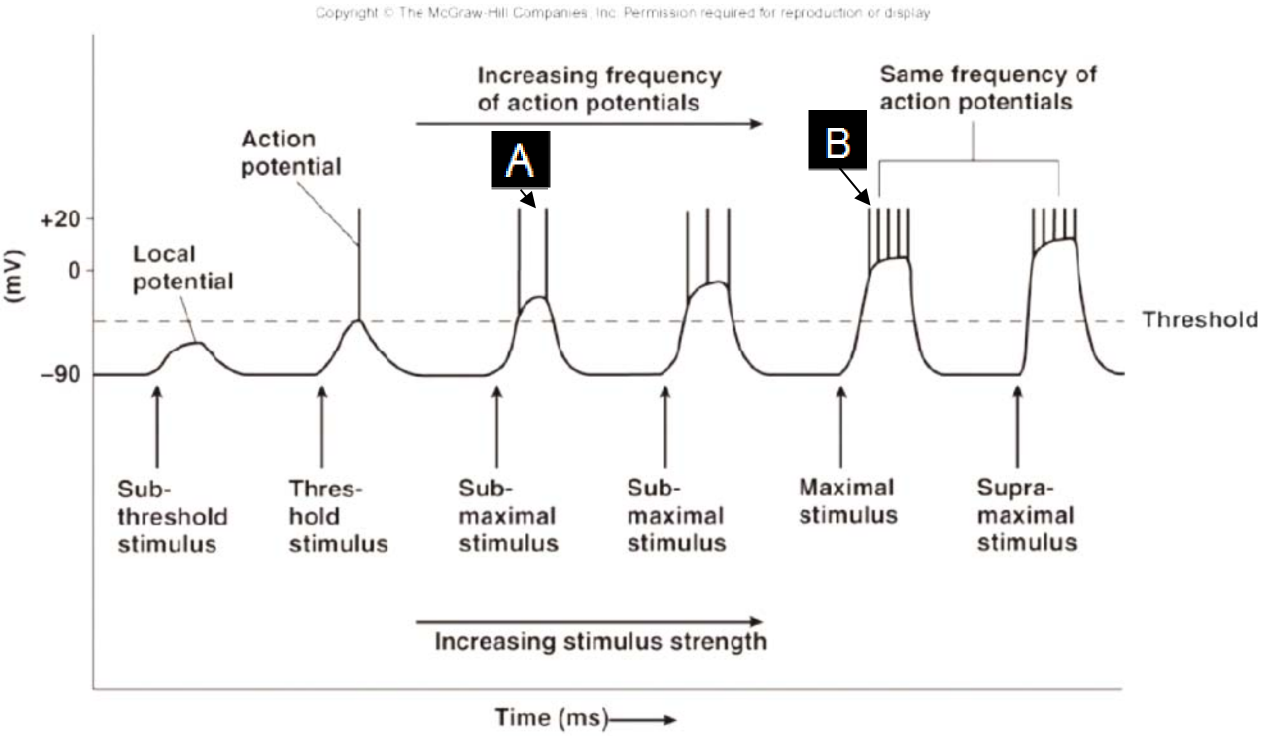

Fig. 7.1

(a) State one factor that contributes to unidirectional conduction of impulses down an axon.

....[1]

With reference to Fig. 7.1 for all questions following,

(b) explain why there is no action potential generated with a sub-threshold stimulus.

....

....

....[2]

(c) explain how the time interval between action potentials seen in label **B** is shorter than the time interval between action potentials seen in label **A**.

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....[3]

(d) state the advantages in the ability of neurons to display these characteristics shown in Fig. 7.1.

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……….[2]

(e) explain what would increase the speed of the signal conduction down the neuron.

………

………

……….[2]

[Total: 10]

8 The Grand Canyon National Park is home to two groups of squirrels. The Albert squirrels *Sciurus aberti* live generally on the south rim of the canyon and the Kaibab squirrels *Sciurus kaibabensis* live on the north rim of the canyon (Fig. 8.1 and Fig. 8.2).

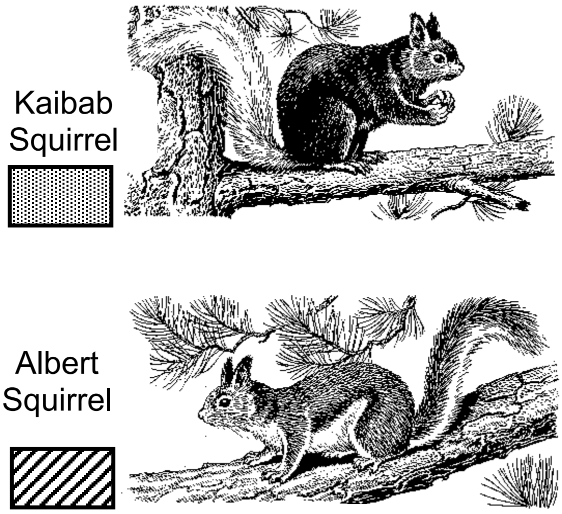

Fig. 8.1

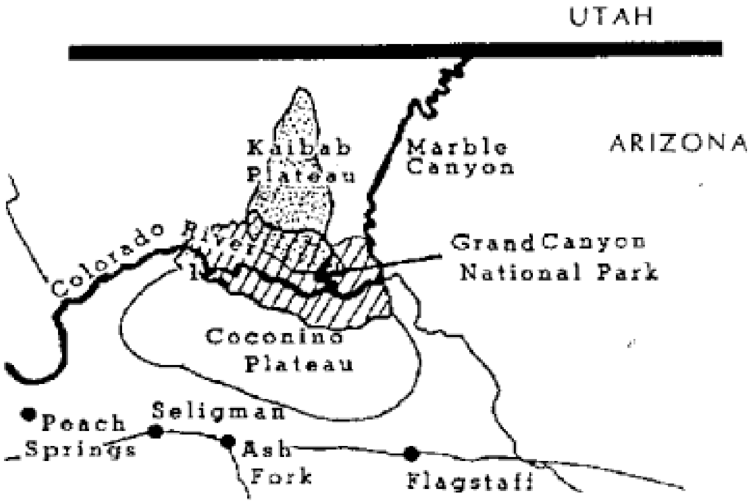

Fig. 8.2

The north rim is about 370 m higher than the south rim. Almost twice as much precipitation falls on the north rim than on the south rim every year. The two groups share many characteristics, but they do not look the same, both groups have tasselled ears, but each group has a unique fur colour pattern.

(a) Explain which Darwinian principles can be applied from the above information on *S. alberti* and *S. kaibabensis*.

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....

....[4]

Several studies have been done on the phylogenetic relationship of the squirrels in and around the Grand Canyon region. Fig. 8.3 is one such study based on cytochrome b DNA sequences.

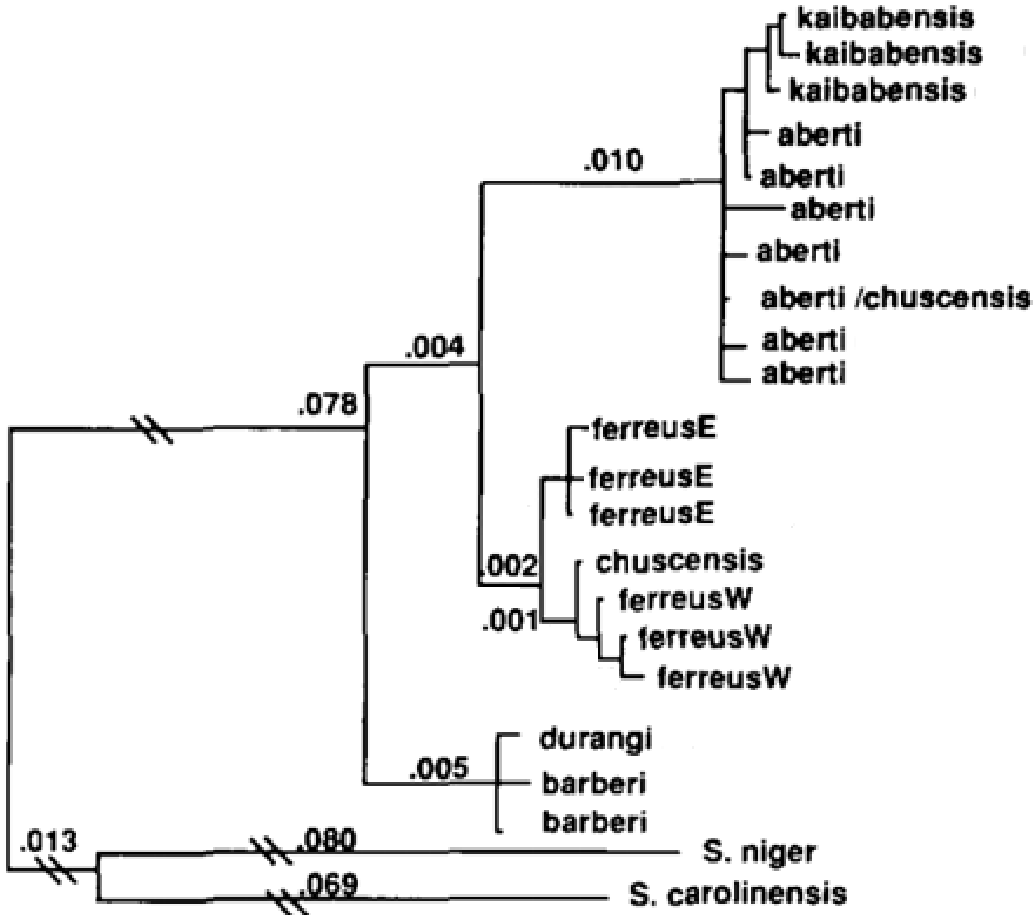

Fig.8.3 Phylogenetic relationship between six *sciurus* subspecies base on cytochrome b sequences constructed by the neighbour-joining method of Saitou and Nei (1987) using the *S. niger* and *S. carolinesis* (Thomas and Martin 1993) sequences as outgroups. Branch lengths and confidence probabilities are noted above and below the branches respectively.

(b) With reference to information already given and also to Fig. 8.3, it is clear that divergent evolution or adaptive radiation is occurring in the evolution of *S. alberti* and *S. kaibabensis*

Explain why it is not convergent evolution.

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……………………………………………………………………………………………………….[2]

(c) **(i)** With reference to information already given and also to Fig. 8.3, suggest with reasons what kind of speciation *S. Alberti* is undergoing.

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……………………………………………………………………………………………………….[2]

(ii) With a known species concept, explain what would be the determining factor confirming *S. alberti* and *S. kaibabensis* as two separate species.

……

……

………[2]

[Total: 10]

Section B

Answer **one** question. Answer each part on a **separate** piece of paper.

Write your answers on separate answer paper provided.
Your answer should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answer must be in continuous prose, where appropriate.
Your answer must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

9 **(a)** Describe cell signaling with the G-protein coupled receptor with a named ligand and its corresponding cellular response. [8]

(b) Explain the advantages and disadvantages of a phosphorylation cascade. [6]

(c) Contrast with elaboration, anaerobic respiration with light independent reaction (Calvin cycle). [6]

[Total: 20]

10 **(a)** Contrast binary fission and mitosis. [6]

(b) Describe generalised and specialised transduction. [8]

(c) In an experiment using T4 bacteriophage, different component molecules were labelled.

- T4 bacteriophages with protein coats labelled with radioactive sulfur.
- T4 bacteriophages with DNA labelled with radioactive phosphorus.

The differently-labelled bacteriophages were allowed to infect host bacteria.
The inside and outside of infected bacteria before lysis were tested for radioactive sulfur and radioactive phosphorus.

Describe and explain the expected results of the experiment. [6]

[Total: 20]

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